

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I RAJESWARI P-192311072 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
RAJESWARI P

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

ANSWERS

1. Reactive Agent-Based Traffic Control System

A Reactive Agent-based Traffic Control System uses an AI agent that responds immediately to real-time traffic conditions. The agent receives data from traffic cameras, sensors, GPS devices, and connected vehicles. The data is processed using streaming technologies such as Apache Kafka and Spark Streaming. The agent follows condition-action rules to control traffic signals. If traffic density is high, it increases the green-light duration, while low traffic results in shorter signal durations. Emergency vehicles can also be given priority. The agent continuously monitors traffic conditions and dynamically changes signal timings to reduce congestion, waiting time, and queue length. The system provides real-time response, automated signal control, improved traffic flow, and efficient emergency management.

2. Goal-Based Agent for Healthcare Diagnosis

A Goal-Based Agent in healthcare makes decisions to achieve specific diagnostic goals. The agent receives patient information such as symptoms, medical history, laboratory results, and medical reports through a data pipeline. It analyzes the collected data using AI models and decision-making rules. The main goals are to identify possible diseases, determine disease severity, select additional diagnostic tests, and identify suitable treatment pathways. Technologies such as Python, PySpark, and databases can process large healthcare datasets efficiently. The agent evaluates different possible decisions and selects the pathway that best satisfies the defined goal. It can provide diagnostic and treatment recommendations for review by healthcare professionals. This system enables faster analysis, personalized decision support, and efficient healthcare data processing.

3. Utility-Based Agent for E-Commerce Recommendations

A Utility-Based Agent generates personalized product recommendations by assigning utility values to different products. The utility value represents how useful or suitable a product is for a particular user. The agent considers factors such as user preferences, previous purchases, product ratings, price, relevance, and availability. Large-scale user interaction data can be processed using PySpark and distributed computing. The agent calculates utility scores for different products and ranks them according to their scores. Products with higher utility values are recommended to the user. User feedback and new purchasing behavior can be continuously analyzed to update the recommendations. This approach improves personalization, user satisfaction, and recommendation accuracy while efficiently processing large amounts of e-commerce data.

4. Learning Agent for Personalized Education System

A Learning Agent is an intelligent agent that improves its performance by learning from user feedback and previous experiences. In an educational platform, the agent analyzes student information such as quiz scores, assignment results, learning time, completed topics, and interaction history. PySpark can process learning data from thousands of students using distributed computing. The agent identifies student strengths and weaknesses and adapts the difficulty of learning materials accordingly. If a student performs poorly, it can provide additional exercises or simpler content. If performance improves, it can gradually provide more challenging content. Feedback from students and their performance is continuously used to improve future recommendations. This creates an adaptive, personalized, and scalable learning environment.

5. Multi-Agent System for Disaster Management

A Multi-Agent System for Disaster Management consists of multiple intelligent agents that collaborate to detect and respond to disasters. Sensor agents collect information from IoT sensors, satellites, weather systems, and other sources. Decision agents analyze the collected data and determine the type, location, and severity of the disaster. Communication agents distribute warnings and emergency information to authorities and affected people. Data can be processed using PySpark and streaming technologies to support real-time analysis. Agents share information and coordinate their actions through communication protocols. When a disaster is detected, decision agents can recommend response strategies while communication agents issue alerts. This distributed approach enables faster detection, coordinated response, efficient resource allocation, and real-time disaster management.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Reactive Agent-Based Traffic Control System

ADD YOUR PRESENTATION PHOTO

1. Introduction

A Reactive Agent-Based Traffic Control System is an intelligent traffic management system that uses agents to observe traffic conditions and immediately take suitable actions. Unlike traditional traffic signals that operate using fixed timings, reactive agents continuously monitor the environment and respond to changes such as traffic congestion, accidents, emergency vehicles, and pedestrian movement.

The main objective is to reduce traffic congestion, minimize waiting time, improve road safety, and provide efficient traffic flow.

2. What is a Reactive Agent?

A reactive agent is an intelligent agent that:

- Observes the current environment.
- Receives information through sensors.

- **Applies predefined rules.**
- **Takes an immediate action.**
- **Does not necessarily maintain a detailed history of previous situations.**

For example:

**IF traffic density is high on Road A
THEN increase the green-light duration for Road A.**

3. System Components

Sensors

- **CCTV cameras**
- **Vehicle detectors**
- **Infrared sensors**
- **GPS information**
- **Emergency vehicle detection systems**

Agent

- **Receives traffic information.**
- **Determines the current traffic condition.**
- **Applies decision rules.**
- **Sends commands to traffic signals.**

Traffic Signal Controller

- **Controls red, yellow, and green signals.**
- **Changes signal timing according to the agent's decision.**

Communication Network

- **Transfers information between intersections and traffic-control agents.**